

Ideation Phase

Empathize & Discover

Date	1 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis Using Tableau
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. It is a useful tool to help teams better understand their users. Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with their goals and challenges.

Empathy Map Canvas (EV Owner/User):

Says "I want a vehicle that's cost-effective in the long run." "I'm worried about the battery performance in extreme weather." "I prefer eco-friendly transportation."	Thinks "Is the charging infrastructure reliable enough for my daily routes?" "Will resale value improve as adoption grows?" "Will I get enough support if there's a tech issue?"
Sees Sees promotions of new EV models from startups and legacy brands. Sees charging stations in metro cities, but gaps in rural areas.	Hears Hears news about government subsidies and EV-friendly regulations. Listens to EV user reviews on social media and YouTube influencers.
Pains Limited charging stations in some areas. Higher upfront cost compared to petrol/diesel vehicles. Concerns about battery life degradation.	Gains Saving fuel cost over time. Environmental benefits and lower emissions. Smooth, noiseless driving experience.

Example Use Case Based on Provided File: User:

Urban commuter in India using an EV for daily office commute. - Scenario: The user starts the day checking their EV battery level via a mobile app. They prefer to charge at night due to lower tariffs. - Pain Point: Midway through their commute, they worry about reaching the office if unexpected detours arise, due to sparse charging spots outside Tier-1 cities. - Gain: The user saves around ₹4,000/month in fuel and appreciates the smooth automatic drive in traffic.

User Experience Flowchart (EV Journey):

Start

→ User researches EV benefits & costs online

- User test-drives and compares EV models
- Purchases EV and downloads the EV companion app
- Plans routes with charging stations shown on app
- Charges EV (home/public)
- Daily commute, app monitors battery + performance
- Receives app alerts for service/battery health
- User shares reviews online
- influences others
- End

Diagram: Empathy Map for EV User

